Mohamed Essam Hashim

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Education

- **University of Medical Sciences & Technology** *Sep 2018 – Apr 2024*
  Bachelor of Medicine, Bachelor of Surgery (M.B.,B.S.) Khartoum, Sudan
  - Graduated at 19 with Distinction / First-Class Honours.
  - Served on the UMST Medical Student Association (UMSA) Executive Board.
  - Thesis: cross-sectional survey of 277 medical students on COVID-19 infection-prevention awareness, attitudes and practices (SPSS); supervised by Dr. Doris Njelekela.

Research Experience

- **Research Fellow — University Hospitals / Case Western Reserve University** *Jun 2026 – Present*
  Quantitative MRI in neurological disease: MR Fingerprinting (MRF) and deep learning in neuro-oncology
  - Developing a deep-learning model to synthesise post-contrast MRI from a 5-minute pre-contrast MRF scan, aiming to reduce gadolinium exposure in repeated neuro-oncology imaging.
  - Member, Case Center for Imaging Research (CCIR).
- **Global Burden of Disease Collaborator — IHME, University of Washington** *Jan 2026 – Present*
  Tools: DisMod-MR, R
  - Reviewing neurological and neuro-oncology burden data for North Africa and the Middle East, including primary data identification and verification for neurovascular pathologies.
  - Appointed to a global network of 11,000+ researchers across 160+ countries quantifying disease burden.
- **Visiting Researcher — Kann Lab, Harvard Medical School / Mass General Brigham** *Sep 2025 – Apr 2026*
  Tools: PyTorch, MONAI, HuggingFace, GitHub
  - Developed LLM-assisted segmentation pipelines for paediatric gliomas (BraTS-PEDs), fusing text and imaging via cross-attention and FiLM with LoRA.
  - Built vision-only baselines (nnU-Net, Swin-UNETR) and designed a synthetic prompt generator for structured imaging outputs.
  - Conducted preprocessing (NIfTI conversion, bias correction, normalisation) and quality control.
  - Drafted figures and methods under mentorship of Dr. Benjamin Kann.
- **Founder — Craniolabs** *May 2025 – Nov 2025*
  Data infrastructure for medical imaging AI in low-resource hospitals
  - Built data-curation pipelines for 300+ CT and MRI studies (de-identification, format conversion, QC).
  - Established data-sharing MoUs with diagnostic centres to source real-world neuroimaging data.
  - Recruited a founding team spanning regulatory (FDA), engineering and clinical backgrounds; admitted to the NVIDIA Inception programme for AI startups.

Research Projects

- **Computational Modelling of Visual Working Memory** *Jun 2026 – Present*
  Advised by Prof. Sanjay Manohar, Nuffield Dept. of Clinical Neurosciences, University of Oxford (remote)
  - Showed that per-individual misbinding weight is not recoverable at realistic trial counts (n=20: sampling SD 0.071 vs. true 0.080; ~30% of fits return zero), redirecting study design to group-level contrasts powered at ~30/arm to detect a 0.05 shift.
  - Established recoverability limits for the three-component (target/misbinding/uniform) mixture model via Monte Carlo simulation of the full generative pipeline; Cramér–Rao bounds placed estimator spread at the theoretical limit, so precision is bounded by trial count, not analysis method.
  - Developing a recurrent neural network study adjudicating persistent-firing versus activity-silent synaptic mechanisms of working-memory maintenance against Oxford Memory Task error patterns.
  - Implementing a published conjunctive-binding architecture (continuous 2D population code), then sweeping plasticity, noise, unit count and inhibition to map model-reachable behavioural error space against clinical phenotypes.

Publications & Preprints

- **Hashim ME**. Blood Interleukin-6 (IL-6) mRNA Expression in Alzheimer's Disease: A Preliminary Analysis. *medRxiv*. Preprint. Posted July 14, 2025. doi:10.1101/2025.07.13.25331470

Clinical Experience

- **El Safa Medical Tower** *May 2024 – Oct 2024*
  Clinical Observer — Neuro-ophthalmology Cairo, Egypt
  - Examined patients incl. Descemet membrane and corneal defects; assisted procedures and aftercare.
  - Took part in multidisciplinary discussions with the attending ophthalmologist.

- **Muhimbili National Hospital (MNH)** *Jun 2023 – Apr 2024*
  Clinical Elective Student — Neurology, Internal Medicine, Surgery Dar es Salaam, Tanzania
  - Examined neurological patients, including cases of progressive multifocal leukoencephalopathy.
  - Joined interdisciplinary rounds (12+ professionals); assisted in surgeries and post-operative care.

- **Rural Residency** *Oct 2022 – Nov 2022*
  Clinical Volunteer Resident Meroe, Sudan
  - Conducted home visits, screenings, counselling and school health education; organised sponsorships, medications and transport for outreach teams.
  - Assessed rural clinics and pharmacies and reported shortages to the Ministry of Health; featured in a media interview on systemic challenges.

Skills

- **Programming:** Python (PyTorch, MONAI, HuggingFace, scikit-learn), R
- **Methods:** Monte Carlo simulation, DisMod-MR
- **Medical Imaging:** DICOM/NIfTI, nnU-Net, Swin-UNETR, ANTs, 3D Slicer
- **Tools:** Git/GitHub, Jupyter, VS Code, Vast.ai GPUs
- **Clinical:** NIHSS, HIPAA/GCP compliance

Certifications

- **CITI Program:** Human Subjects Research (HSR) – Biomedical Research *May 2025*
- **National Institute on Drug Abuse:** Good Clinical Practice (GCP) *Apr 2025*
- **National Institutes of Health:** NIH Stroke Scale Certification (NIHSS) *Aug 2024*